

Experiment No : 01

Experiment Name : Study of I-V characteristics of a p-n junction diode

Objectives:

- To observe and draw the forward and reverse bias I-V characteristics of a p-n junction diode
- To calculate static and dynamic resistance in both forward and reverse bias condition

Theory:

A p-n junction diode is created by combining p-type and n-type semiconductor regions. Its I-V characteristic curve shows how the current through the diode changes with the applied voltage, helping explain its operation under different biasing conditions.

Biasing of the junction:

- Forward Bias: Connecting the p-side to the positive terminal and n-side to the negative terminal lowers the junction barrier. Once the applied voltage becomes high enough (0.7V for Si, 0.3 for Ge), a large current begins to flow.
- Reverse Bias: Reversing the terminals increases the barrier, blocking majority carriers. Only a tiny leakage current from minority carrier flows. At high reverse voltages, breakdown can occur (Zener or Avalanche) which causes a sudden rise in current

I-V Characteristics:

The diode current follows the Shockly equation,

$$I_D = I_S \left(e^{\frac{V}{nV_T}} - 1 \right)$$

Where,

I_D = Diode Current

I_S = Reverse saturation current

V = Applied voltage

n = Ideality factor

V_T = Thermal voltage (25.9 mV at room temperature)

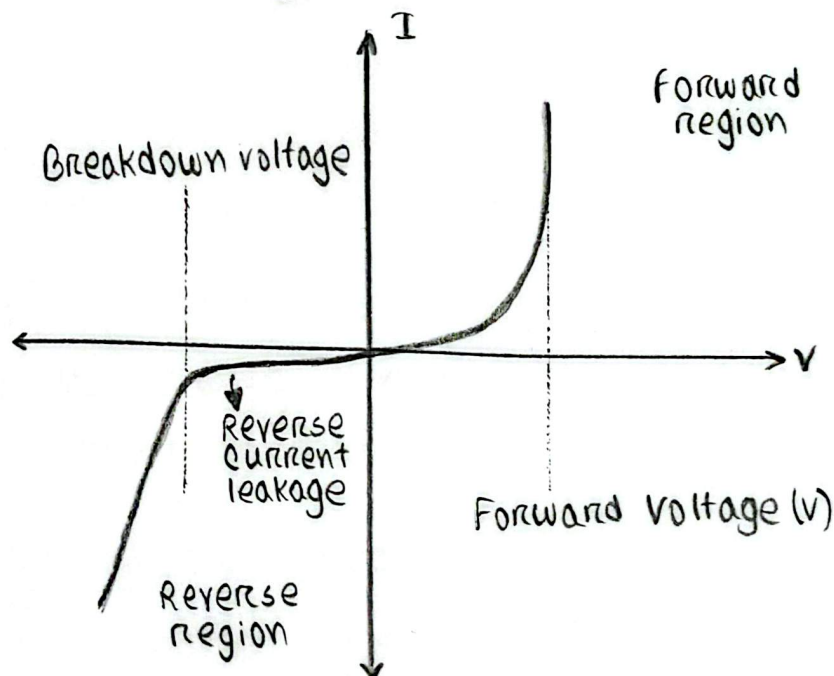


Figure: I-V characteristics of a p-n junction diode

Required Apparatus:

- Diode (1N4007, 1 piece)
- Resistor (1 k Ω , 1 piece)
- DC supply (0-15 V)
- Ammeter (0-2 A)
- Voltmeter (0-5 V)
- Project Board
- Connecting wires

Circuit Diagram:

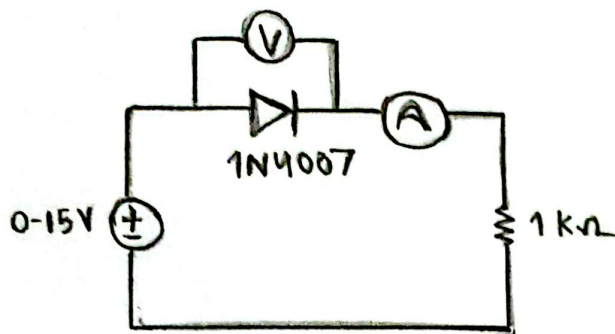


Figure: Circuit Diagram for determining I-V characteristics at forward bias

Procedure:

- (i) The diode was connected in forward bias by linking the p-terminal to the positive and the n-terminal to the negative of the supply as shown in the above circuit diagram
- (ii) The applied voltage was increased slowly and the corresponding current was measured with the ammeter at each step
- (iii) Readings were recorded until noticeable conduction was observed, typically near 0.7V for a Si diode.
- (iv) Then the Voltage (V) vs Current (I) were plotted on a graph for forward bias

Data Table:

Source Voltage, V_s (V)	Diode Voltage, V_D (V)	Diode Current, I_D (mA)
1.01	0.49	0
3.31	0.60	5
7.53	0.68	11
8.61	0.69	14
10.93	0.72	18
12.38	0.74	21
15.30	0.76	26

Discussion and Conclusion:

The plotted I-V curve was close to the expected shape, although the ideal exponential rise in forward bias was not fully observed due to practical limitations such as temperature variation and measurement obstacles. Despite these small deviations, the overall behaviour of the diode matched theoretical principles.

Precautions:

- All electrical devices were connected correctly
- Avoided parallax error while taking readings from analog meters